

Skateboard Deck Design

Assignment 2
ENG ME 360: Product Design
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1 Introduction

The main objective of this assignment was to design a lightweight deck of a skateboard such that it will meet specific criteria as indicated below. First, a numerical analysis will be done, in which the max displacement will be calculated. In order to test the skateboard's strength and max displacement, a model of the skateboard deck will be created using SOLIDWORKS, and a Finite Element Analysis (FEA) will be used to determine the max displacement, stress and strain on the skateboard deck. The following is the criteria that this skateboard deck should meet:

- Maximum vertical displacement at any point on the skateboard deck should be 0.375 inches.
- Factor of safety of 3 should be accounted for, meaning, the maximum stress on the skateboard should be a third of the yield stress of the material being used for the deck.
- Maximum weight that is applied on the deck should be 180 lbs.

2 Material Selection

The first decision to make was, what type of material should be used: metal, plastic, or wood? At the end, wood was chosen. This was essentially because plastic have a chance of melting if used for too long outside or left outside. Moreover, metals have a great thermal conductivity, which means that the skateboard could heat up and injure the user. However, with wood, it's not only an insulator, thus, avoiding any chance of injuring the user, but it also will not melt if left outside for long periods of time. For this reason, wood was chosen as the material of the skateboard deck.

To figure out what kind of wood to be chosen, the "Mechanical Properties of Wood" manual was used. The goal of this assignment was to minimize deflection, minimize weight, and maximize the yield stress such that the skateboard deck will not plastically deform upon a weight applied onto it. The following are the three best kinds of wood determined from the manual for the specifics of the skateboard deck.

- Maple Wood:
 - Density: 500 kg/m^3
 - Fibre Stress at Elastic Limit: 85 MPa
 - Modulus of Elasticity: 10 GPa
- Oak Wood -
 - Density: 640 kg/m^3
 - Fibre Stress at Elastic Limit: 100 MPa
 - Modulus of Elasticity: 12 GPa
- Pine Wood -
 - Density: 420 kg/m^3
 - Fibre Stress at Elastic Limit: 80 MPa
 - Modulus of Elasticity: 11 GPa

The above list shows the materials that have a high fibre stress at elastic limit, high modulus of elasticity, and a low density. Among the materials above, Oak was chosen to be the material of the skateboard deck because it has the highest of the fibre stress at elastic limit, which means that more force should be applied to an Oak skateboard to plastically deform it than that for skateboards made out of Maple and Pine wood. Moreover, Oak wood also has the highest Modulus of Elasticity, which means that the Oak will need more stress to be applied onto itself to bend it than Pine and Maple wood. Thus, having a high modulus of elasticity will help decrease the vertical displacement. On the other hand, Oak wood does have a higher density, but this will not make too much of difference, since the volume of the skateboard is very small, so the mass difference between the three kinds of wood will not be significant.

3 Dimension of the Skateboard Deck

The goal of the design was to create a standard skateboard, which meant that some of the essential dimensions of the skateboard will be standardized. This is because typical consumer of this product will require to have a certain specific set of dimensions in order to fulfill their needs with the product. For this reason, the length and the width of the skateboard deck was standardized and found to be the following.

$$\text{Length} = 32\text{in} = 800\text{mm} \quad (1)$$

$$\text{Width} = 8\text{in} = 203\text{mm} \quad (2)$$

Thickness of the skateboard was not standardized, rather determined using Mechanics of Materials.

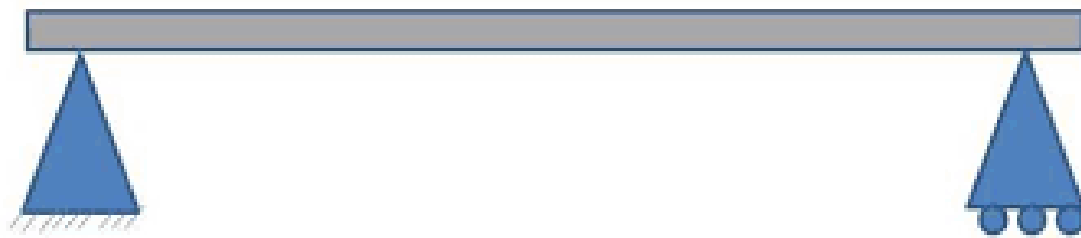
4 Assumptions

In order to complete a numerical analysis, a few assumptions will be made. The following are the assumptions used for the numerical computation.

- The cross-section of the skateboard deck will be rectangular.
- The load applied on the deck will be on the center of the deck since this accounts for the greatest vertical displacement.
- A pin and a roller joint will be considered on the trucks of the skateboard deck, since this will be the appropriate system to test for vertical displacement.

5 Numerical Analysis

The following image shows a simplified version of the system that is analogous to the skateboard deck.



Simply Supported Beam

Figure 1: Diagram of a Beam on a Pin and Roller Joints

The following is the derivation of finding the thickness of the skateboard such that constraint for maximum stress is met.

$$\text{(For Rectangular Cross-Section)} \quad (3)$$

$$\sigma_{max} = \frac{6M}{bh^2} \quad (4)$$

$$\text{(Moment at the center of the beam where stress will be maximum)} \quad (5)$$

$$M = \frac{PL}{4} \quad (6)$$

$$\Rightarrow \sigma_{max} = \frac{6PL}{4bh^2} < \frac{\sigma_{yield}}{\text{Factor of Safety}} = \frac{100MPa}{3} \quad (7)$$

$$\Rightarrow h^2 > \frac{18PL}{4b\sigma_{yield}} \quad (8)$$

$$P = 180\text{lbf} = 800N \quad (9)$$

$$L_{\text{trunk-to-trunk}} = 20\text{in} = 0.5m \quad (10)$$

$$b = 8\text{in} = 0.2m \quad (11)$$

$$\Rightarrow h > 0.0095m \Rightarrow h > 9.5mm \quad (12)$$

The above shows that the thickness of the skateboard should be at least 9.5 mm such that maximum stress on the deck will be a third of the yield stress of Oak wood.

The following is the derivation to find minimum height such that the maximum vertical displacement constraint is met.

$$v_{max} = 0.375\text{in} \quad (13)$$

$$v_{max} > \frac{PL^3}{48EI} \quad (14)$$

$$\Rightarrow v_{max} > \frac{12PL^3}{48Ebh^3} \quad (15)$$

$$\Rightarrow h > \left(\frac{12PL^3}{48Ebv_{max}} \right)^{1/3} \quad (16)$$

$$\Rightarrow h > 0.010m = 10mm \quad (17)$$

The above shows that the minimum thickness of the skateboard deck in order to constraint the maximum vertical displacement of 0.375 inches. From the above, we see that the minimum thickness should be 10 mm.

From the above computation, we can see that in order for the skateboard deck to satisfy both constraints: maximum vertical displacement and maximum stress, the thickness of the deck should at least be 10 mm. To be a little bit more cautious, it was decided to make the thickness of the skateboard 13mm just to account for any other unforeseen forces.

6 SOLIDWORKS Model



Figure 2: Image of the Skateboard Model

The above image is the SOLIDWORKS model of the skateboard deck. The following is a quick overview of how the SOLIDWORKS model was created. The mass of the skateboard deck was calculated to be 1.3 kg.

6.1 Sweep

First, a drawing of the side profile was created and swept on a curved trajectory to get the contour of the skateboard.

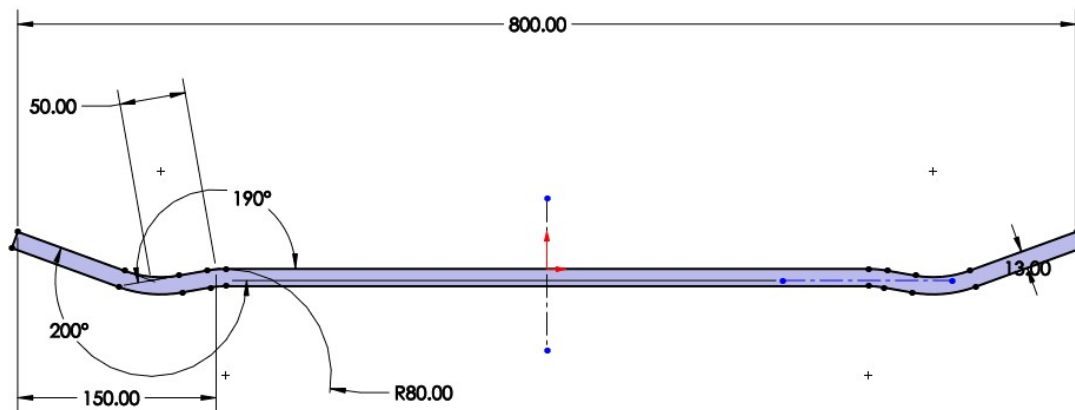


Figure 3: Image of the Side Profile of the Skateboard

6.2 Fillet

Finally, the fillets were added to the sides to get the curved look of the skateboard.

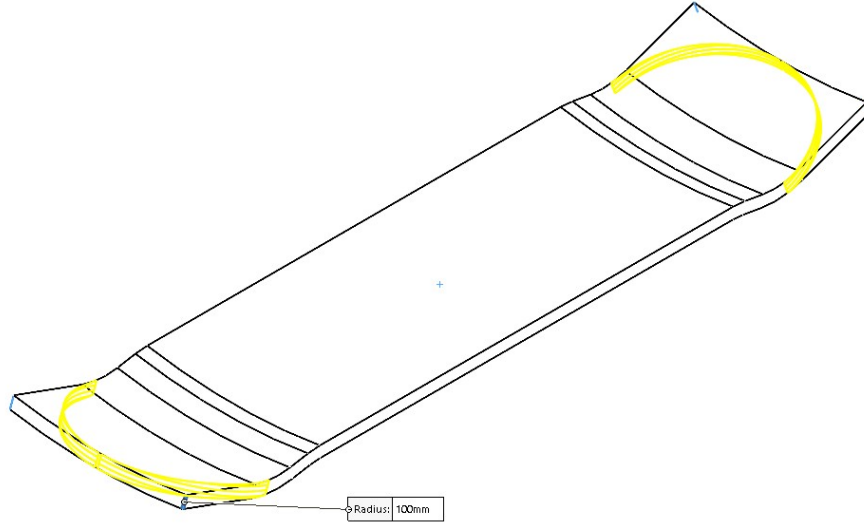


Figure 4: Fillets on the Side of the Skateboard

7 Simulation

Before starting the simulation, few split lines were created so that we could simulate the weight being applied onto the skateboard and the trunks acting as a pin and roller joints.

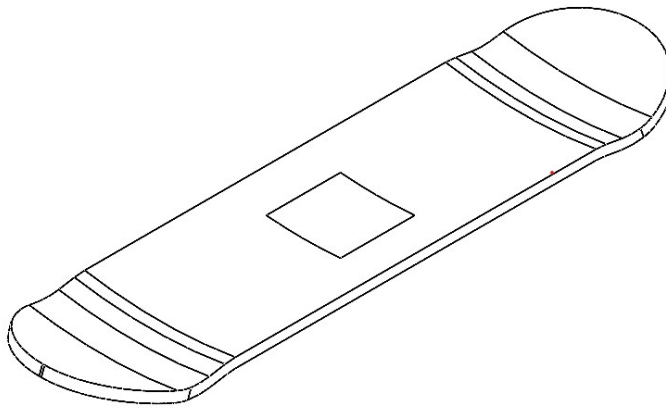


Figure 5: Image of the Skateboard Model with Split Lines

The small square seen on the skateboard is the split line used to simulate the area where the force is being applied. Moreover, the area of that split line is the same as the surface area of a US 12 shoe.

7.1 Displacement Simulation

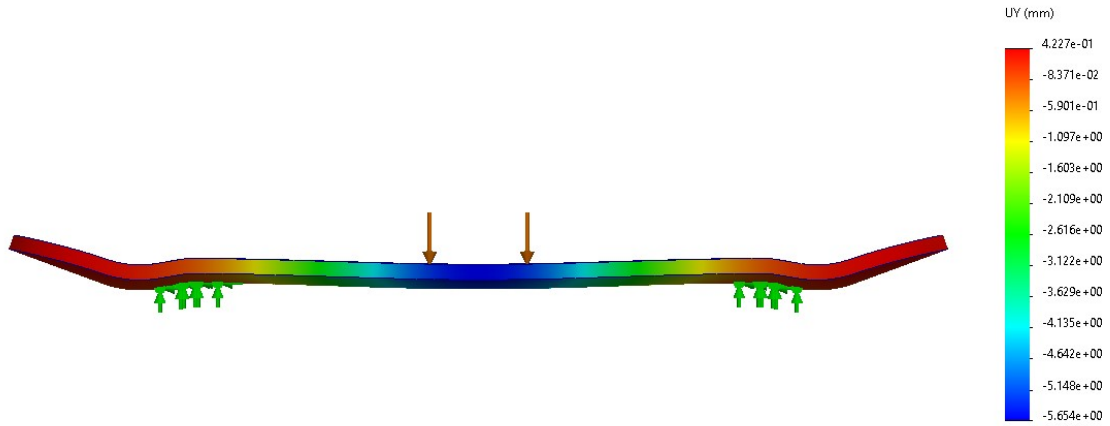


Figure 6: Results from Displacement Simulation

As seen above in the displacement graph, the maximum displacement of the skateboard was only about 5.7 mm, which is only 0.22 inches. This shows that the maximum vertical displacement constraint of 0.375 is met.

7.2 Stress Simulation

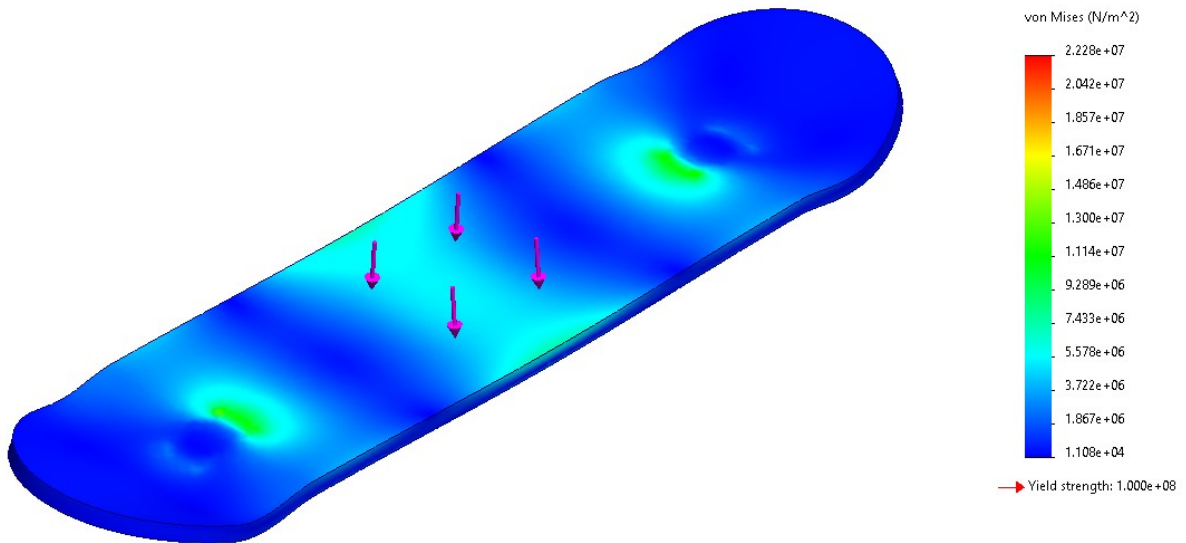


Figure 7: Results from a Stress Simulation

The maximum stress on the skateboard deck was 22 MPa, which is less than a third of the yield stress ($\frac{yieldstress}{3} = 33.3MPa$). This shows that the stress constraint of maximum stress being less than a third of the yield stress is met.

8 Conclusion and Future Work

As seen in the previous sections, we can see that the FEA tool is very useful in designing a product and ensuring that the constraints of the product is met. Moreover, it needs to be stated that the results of the FEA tool is not perfect, especially because the loads and fixtures applied on the skateboard deck were not purely realistic. The future work for this assignment includes decreasing the thickness in order to decrease the mass of the product and conducting another FEA to ensure that the constraints are still met. Moreover, new skateboard designs will be tested to see how different designs will effects the mass, the maximum vertical displacement, and the maximum stress.